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Hierarchical Algorithms
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Exercise sheet 04, Solution hints
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Homework:

Exercise 1: For the calculation of an LU decomposition of an $\mathcal{H}_p$-matrix with $p > 1$, we need a (recursively) formatted forward substitution where we compute a matrix $B \in \mathcal{R}_{p-1}$ with an equation of the form $A = LB$ with given $A \in \mathcal{R}_{p-1}$ and normalized lower triangular matrix $L = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \in \mathcal{H}_{p-1}$. The matrices $L_{21} \in \mathcal{R}_{p-2}$, and $L_{11}, L_{22} \in \mathcal{H}_{p-2}$ are normalized lower triangular matrices with the same structure as L.

a) Think about how the (recursively defined) formatted forward substitution is performed.

b) Show that the computational cost $\mathcal{N}_{\text{fsubst}}(p-1)$ is given by

$$\mathcal{N}_{fsubst}(p-1) = n \log_2 n - \frac{3n}{2} + 1$$

Solution hints:

a) Let $a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \in \mathbb{R}^{2^{p-1}}$ be the factors of the matrix $A \in \mathcal{R}_{p-1}$ with $a_1, a_2, b_1, b_2 \in \mathbb{R}^{2^{p-2}}$, i.e. we have $ab^T = A$. We obtain

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = LB$$

with $A_{ij} = a_i b_j^T, B_{ij} \in \mathcal{R}_{p-2}$ for $i, j \in \{1, 2\}$. For the computation of $B \in \mathcal{R}_{p-1}$, we need factors $c = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}, d = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \in \mathbb{R}^{2^{p-1}}$ with $c_1, c_2, d_1, d_2 \in \mathbb{R}^{2^{p-2}}$ such that $cd^T = B$. With $d := b$, we obtain $A = LB \iff ab^T = L(cb^T) = (Lc)b^T$. So, we determine c by solving $a = Lc$:

$$a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = Lc.$$

b) The calculation of B in $A = LB$ requires only the solution of $a = Lc$. We obtain the solution with the following steps:

(i) Compute c_1 recursively with $a_1 = L_{11}c_1$ with the costs $\mathcal{N}_{\text{fsubst}}(p-2)$.

(ii) Compute c_2 recursively with $a_2 = L_{21}c_1 + L_{22}c_2$, i.e. $L_{22}c_2 = a_2 - L_{21}c_1$, with the costs
$\underbrace{3 \cdot \frac{n}{4} - 1}_{\text{for } L_{21}c_1} + \frac{n}{4} + \mathcal{N}_{\text{fsubst}}(p-2) = \mathcal{N}_{\text{fsubst}}(p-2) + n - 1.$

For $p > 1$, we obtain

$$\mathcal{N}_{\text{fsubst}}(p-1) = 2\mathcal{N}_{\text{fsubst}}(p-2) + n - 1 = 2\mathcal{N}_{\text{fsubst}}(p-2) + 2^p - 1$$

and for $p = 0$ we have $\mathcal{N}_{\text{fsubst}}(0) = 0$ since $A = LB$ simplifies to $A = B$ as $L = 1 \in \mathbb{R}^{1\times 1}$. The costs amount to

$$\begin{aligned}
\mathcal{N}_{\text{fsubst}}(p-1) &= 2\mathcal{N}_{\text{fsubst}}(p-2) + 2^p - 1 \\
&\overset{p>2}{=} 2(2\mathcal{N}_{\text{fsubst}}(p-3) + 2^{p-1} - 1) + 2^p - 1 \\
&\overset{p>2}{=} 2^2\mathcal{N}_{\text{fsubst}}(p-3) + (2 \cdot 2^p) - (2+1) \\
&= \cdots \\
&= 2^{p-1}\mathcal{N}_{\text{fsubst}}(0) + 2^p(p-1) - \sum_{i=0}^{p-2} 2^i \\
&= 2^{p-1} \cdot 0 + 2^p \cdot p - 2^p - \frac{1 - 2^{p-2+1}}{1-2} \\
&= n \log_2 n - n + 1 - \frac{n}{2} \\
&= n \log_2 n - \frac{3n}{2} + 1.
\end{aligned}$$

We hence obtain $\mathcal{N}_{\text{fsubst}}(p-1) = \mathcal{O}(n \log n)$.

Exercise 2: A series of matrices $A_k \in \mathbb{R}^{m\times n}$, $k \in \mathbb{N}$, converges to the limit

$$\lim_{k\to\infty} A_k = A \in \mathbb{R}^{m\times n}.$$

Show that a $k_0 \in \mathbb{N}$ exists with $\operatorname{rank}(A) \le \operatorname{rank}(A_k)$ for all $k \ge k_0$.

Solution hints: Let $r = \operatorname{rank}(A) \le \min\{m, n\}$. Then, there exist indices

$$\sigma = \{i_1, \dots, i_r\} \subseteq \{1, \dots, m\} \text{ and } \tau = \{i_1, \dots, i_r\} \subseteq \{1, \dots, n\},$$

such that the submatrix

$$A|_{\sigma\times\tau} = (a_{i,j})_{i\in\sigma, j\in\tau} \in \mathbb{R}^{r\times r}$$

is nonsingular. The set of all invertible matrices is open since the determinant function is continuous, the set $\mathbb{R} \setminus \{0\}$ is open and preimages of open sets with continuous mappings are open. Hence, there exists a $k_0 \in \mathbb{N}$ with nonsingular $A_k|_{\sigma\times\tau}$ for all $k \ge k_0$. The claim follows since nonsingular matrices have full rank.
An alternative proof is based on the singular value decomposition of A. Let σ_r be the smallest

positive singular value of A. Then there exist orthogonal matrices $U \in \mathbb{R}^{m\times m}, V \in \mathbb{R}^{n\times n}$ with

$$A = U\Sigma V^T \text{ and } \Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \ddots & & & & \\ & & \sigma_r & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix} \in \mathbb{R}^{m\times n}.$$

Hence, $\operatorname{rank}(A) = r \leq \min\{m, n\}$. Furthermore, we have

$$\lim_{k\to\infty} A_k = A \Leftrightarrow \lim_{k\to\infty} \|A_k - A\|_2 = 0,$$

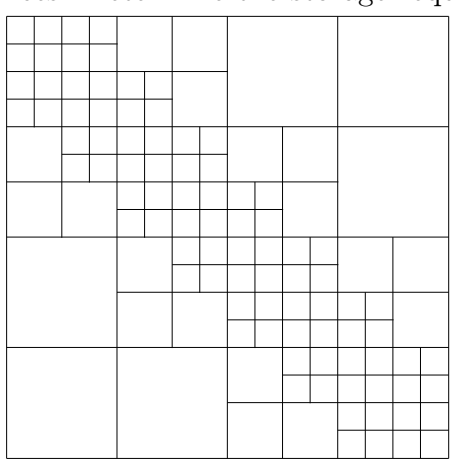
since all norms are equivalent in finite dimensional vector spaces. Furthermore, we have

$$\min_{\operatorname{rank}(\tilde{A})<r} \|A - \tilde{A}\|_2 = \sigma_r.$$

For $\varepsilon := \frac{\sigma_r}{2}$ thus holds $\|A - \tilde{A}\|_2 > \varepsilon$ for all $\tilde{A}$ with $\operatorname{rank}(\tilde{A}) < r$, and it exists a $k_0 \in \mathbb{N}$ with $\|A_k - A\|_2 < \varepsilon$ for all $k \geq k_0$. So, the statement follows.

Exercises (in class):

Exercise 3: We refine the $\mathcal{H}_p$-matrix format by subdividing not only the diagonal matrix blocks but also those blocks that are adjacent to the block diagonal, i.e. we refine any block that contains an index pair (i, j) with $|i - j| \leq 1$. A typical structure is shown below for a matrix in $\mathbb{R}^{16\times 16}$. Any matrix blocks with at least two rows and columns are filled with $\mathcal{R}(1)$ matrices. Determine the storage requirement for a matrix of this type with 2^p rows and columns.



Solution hints: The matrix format is given by

$$\mathcal{H} = \left(\begin{array}{c|c} \mathcal{H} & \mathcal{U} \\ \hline \mathcal{U}^{\mathrm{T}} & \mathcal{H} \end{array}\right) \qquad \text{with} \qquad \mathcal{U} = \left(\begin{array}{c|c} \mathcal{R} & \mathcal{R} \\ \hline \mathcal{U} & \mathcal{R} \end{array}\right)$$

Assumption: The memory requirements for $n = 2^p$ are

$$\begin{aligned} S_{\mathcal{H}}(p) &= 6p2^p - 12 \cdot 2^p + 16 \\ &= 6(\log_2 n)n - 12n + 16 \end{aligned}$$

where we assume that $S_{\mathcal{H}}(1) = S_{\mathcal{U}}(1) = 4$.
Derivation of the assumption: The memory requirements of $\mathcal{U}$ are given by

$$\begin{aligned}
S_{\mathcal{U}}(p) &= S_{\mathcal{U}}(p-1) + 3S_{\mathcal{R}}(p-1) \\
&= S_{\mathcal{U}}(p-1) + 3 \cdot 2 \cdot 2^{p-1} \\
&= S_{\mathcal{U}}(p-2) + 3 \cdot 2^{p-1} + 3 \cdot 2^{p} \\
&= S_{\mathcal{U}}(p-2) + 3 \cdot (2^{p-1} + 2^{p}) \\
&= \ldots = S_{\mathcal{U}}(1) + 3 \cdot (2^{1} + \ldots + 2^{p}) \\
&= 4 + 3 \cdot (2^{p-1} - 1) \\
&= 3 \cdot 2^{p-1} - 8
\end{aligned}$$

The memory requirements of $\mathcal{H}$ are given by

$$\begin{aligned}
S_{\mathcal{H}}(p) &= 2S_{\mathcal{H}}(p-1) + 2S_{\mathcal{U}}(p-1) \\
&= 2S_{\mathcal{H}}(p-1) + 2(3 \cdot 2^{p} - 8) \\
&= 2[2S_{\mathcal{H}}(p-2) + 2(3 \cdot 2^{p-1} - 8)] + 2(3 \cdot 2^{p} - 8) \\
&= 2^2 S_{\mathcal{H}}(p-2) + \underbrace{3 \cdot 2^{p+1} - 2^2 \cdot 8 + 3 \cdot 2^{p+1} - 2 \cdot 8}_{=2 \cdot 3 \cdot 2^{p+1} - 8(2^2+2)} \\
&= \ldots \\
&= 2^{p+1} S_{\mathcal{H}}(1) + (p-1) \cdot 3 \cdot 2^{p+1} - 8(2 + 2^2 + \ldots + 2^{p+1}) \\
&= 2^{p+1} + (p-1) \cdot 3 \cdot 2^{p+1} - 8(2^{p} - 2) \\
&= 3 \cdot p \cdot 2^{p+1} - 2 \cdot 2^{p+1} - 4 \cdot 2^{p+1} + 16 \\
&= 6p2^{p} - 12 \cdot 2^{p} + 16
\end{aligned}$$

Proof per induction for $S_{\mathcal{U}}(p)$:

Induction hypothesis: $S_{\mathcal{U}}(p) = 3 \cdot 2^{p+1} - 8$

Statement: $S_{\mathcal{U}}(p+1) = 3 \cdot 2^{p+2} - 8$

Proof:

$$\begin{aligned}
S_{\mathcal{U}}(p+1) &= S_{\mathcal{U}}(p) + 3S_{\mathcal{R}}(p) \\
&= 3 \cdot 2^{p+1} - 8 + 3 \cdot 2 \cdot 2^{p} \\
&= 3 \cdot 2^{p+2} - 8
\end{aligned}$$

Proof per induction for $S_{\mathcal{H}}(p)$:

Induction hypothesis: $S_{\mathcal{H}}(p) = 6p \cdot 2^{p} - 12 \cdot 2^{p} + 16$

Statement: $S_{\mathcal{H}}(p+1) = 6(p+1) \cdot 2^{p+1} - 12 \cdot 2^{p+1} + 16$

Proof:

$$\begin{aligned}
S_{\mathcal{H}}(p+1) &= 2S_{\mathcal{H}}(p) + 2S_{\mathcal{U}}(p) \\
&= 2[6p \cdot 2^{p} - 12 \cdot 2^{p} + 16] + 2[3 \cdot 2^{p+1} - 8] \\
&= 6p2^{p+1} - 12 \cdot 2^{p+1} + 32 + 6 \cdot 2^{p+1} - 16 \\
&= 6(p+1)2^{p+1} - 12 \cdot 2^{p+1} + 16
\end{aligned}$$

Exercise 4: The following pseudocode shows a possible implementation of the formatted factorization of an $\mathcal{H}_p$-matrix.

Input: $\mathcal{H}_p$-matrix H with $p = 0$ or $H = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}$.

Output: $\mathcal{H}_p$-matrix Factors L and U with $H \approx LU$

1: **function** `lu_decomposition_hp_matrix`(H)
2: **if** p > 0 **then**
3: $L_{11}, U_{11} \leftarrow$ `lu_decomposition_hp_matrix`(H_{11})
4: $L_{21} \leftarrow H_{21} U_{11}^{-1}$
5: $U_{12} \leftarrow L_{11}^{-1} H_{12}$
6: $H_{22} \leftarrow H_{22} \ominus L_{21} \odot U_{12}$
7: $L_{22}, U_{22} \leftarrow$ `lu_decomposition_hp_matrix`(H_{22})
8: $L \leftarrow \begin{bmatrix} L_{11} & \\ L_{21} & L_{22} \end{bmatrix}$
9: $U \leftarrow \begin{bmatrix} U_{11} & U_{12} \\ & U_{22} \end{bmatrix}$
10: **else**
11: $L \leftarrow (1)$
12: $U \leftarrow H$
13: **end if**
14: **return** L, U
15: **end function**

Show that the computational cost for the (formatted) LU factorization of an $\mathcal{H}_p$-matrix is given by

$$\mathcal{N}_{\mathrm{LU}}(p) = \frac{21}{4} n \log_2^2 n + \frac{61}{4} n \log_2 n - 37(n-1).$$

<u>**Hint:**</u> The cost for the solves with triangular matrices in lines 4 and 5 of the algorithm can both be estimated by the result from exercise 1. Note that the backward substitution requires one additional division for $p = 0$ compared to the forward substitution.

Solution hints: The LU factorization as shown above requires

1. Two recursive computations of the (formatted) LU factorization costing $\mathcal{N}_{\mathrm{LU}}(p-1)$.

2. Two triangular solves costing

$$\mathcal{N}_{\mathrm{fsubst}}(p-1) = n \log_2 n - \frac{3}{2} n + 1,$$

and

$$\mathcal{N}_{bsubst}(p-1) = n \log_2 n - n + 1.$$

3. One multiplication of rank-1 matrices costing

$$\mathcal{N}_{\mathcal{RR}}(p-1) = 3\frac{n}{2} - 1.$$

4. One subtraction of a rank-1 matrix from a $\mathcal{H}_{p-1}$ matrix costing

$$\mathcal{N}_{\mathcal{H}+\mathcal{R}}(p-1) = 17\frac{n}{2}\log_2\frac{n}{2} + 39\frac{n}{2} - 38.$$

This yields

$$\begin{aligned}
\mathcal{N}_{\mathrm{LU}}(p) =& 2\mathcal{N}_{\mathrm{LU}}(p-1) + \mathcal{N}_{\mathrm{fsubst}}(p-1) + \mathcal{N}_{\mathrm{bsubst}}(p-1) + \mathcal{N}_{\mathcal{RR}}(p-1) + \mathcal{N}_{\mathcal{H}+\mathcal{R}}(p-1)\\
=& 2\mathcal{N}_{\mathrm{LU}}(p-1) + 2n\log_2 n - \frac{5}{2}n + 2 + \frac{3}{2}n - 1 + \frac{17}{2}n\log_2\frac{n}{2} + \frac{39}{2}n - 38\\
=& 2\mathcal{N}_{\mathrm{LU}}(p-1) + 2np - \frac{5}{2}n + 2 + \frac{3}{2}n - 1 + \frac{17}{2}n(p-1) + \frac{39}{2}n - 38\\
=& 2\mathcal{N}_{\mathrm{LU}}(p-1) + (2+\frac{17}{2})np + (-\frac{5}{2} + \frac{3}{2} - \frac{17}{2} + \frac{39}{2}n) - 37\\
=& 2\mathcal{N}_{\mathrm{LU}}(p-1) + \frac{21}{2}np + \frac{20}{2}n - 37\\
=& 2\left(2\mathcal{N}_{\mathrm{LU}}(p-2) + \frac{21}{2}\frac{n}{2}(p-1) + \frac{20}{2}\frac{n}{2} - 37\right) + \frac{21}{2}np + \frac{20}{2}n - 37\\
=& 2^2\mathcal{N}_{\mathrm{LU}}(p-2) + 2\frac{21}{2}np + 2\frac{19}{2}n - \frac{21}{2}n - (1+2)37\\
=& 2^2\left(2\mathcal{N}_{\mathrm{LU}}(p-3) + \frac{21}{2^2}\frac{n}{2^2}(p-2) + \frac{20}{2}\frac{n}{2^2} - 37\right)\\
&+ 2\frac{21}{2}np + 2\frac{20}{2}n - \frac{21}{2}n - (1+2)37\\
=& 2^3\mathcal{N}_{\mathrm{LU}}(p-3) + 3\frac{21}{2}np + 3\frac{20}{2}n - (1+2)\frac{21}{2}n - (1+2+2^2)37\\
=& \cdots = 2^p\mathcal{N}_{\mathrm{LU}}(0) + p\frac{21}{2}np + p\frac{20}{2}n - \left(\sum_{k=1}^{p-1}k\right)\frac{21}{2}n - \left(\sum_{k=1}^{p-1}2^k\right)\\
=& n\mathcal{N}_{\mathrm{LU}}(0) + \frac{21}{2}np^2 + \frac{20}{2}np - \frac{p(p-1)}{2}\frac{21}{2}n - (2^p-1)37\\
=& n\mathcal{N}_{\mathrm{LU}}(0) + \frac{21}{2}np^2 - \frac{21}{4}np^2 + \frac{20}{2}np + \frac{21}{4}np - (n-1)37\\
=& n\mathcal{N}_{\mathrm{LU}}(0) + \frac{21}{4}np^2 + \frac{61}{4}np - (n-1)37.
\end{aligned}$$

Since the algorithm does not compute anything in the case $p = 0$, we have $\mathcal{N}_{\mathrm{LU}}(0) = 0$ and therefore

$$\mathcal{N}_{\mathrm{LU}}(p) = \frac{21}{4}np^2 + \frac{61}{4}np - (n-1)37.$$